AGROLOOP — BUSINESS MODEL CANVAS & UNIT ECONOMICS

Course: Design Experience III (AY 2026–27) | **Student:** Veer Nagda | **Mentor:** Prof. Deepali A. Maste | **Vertical:** Entrepreneurship

1. The Nine-Block Business Model Canvas Matrix

<table>
<tr>
<td>1. KEY PARTNERS

• MCGM & TMC Sanitation Depts
• Gated Societies (Powai, Bandra)
• Palghar Farmer Producer Orgs
• EV Fleet Providers (Euler)
• KVK Palghar & AIC-NMIMS</td>
<td>2. KEY ACTIVITIES

• Scheduled EV scrap harvesting
• Doorstep optical purity grading
• 4-Stream biological conversion
• Gemini AI agronomist advisory
• SWM compliance certification</td>
<td rowspan="2">3. VALUE PROPOSITIONS

For Upstream (City):
• 100% SWM 2016 compliance
• Stench-free daily morning pickup
• Digital audit-ready ESG certs

For Downstream (Farms):
• 35% cheaper inputs vs chemical
• +28% soil water retention
• Vernacular AI crop advisory</td>
<td>4. CUSTOMER RELATIONS

• Dedicated B2B Key Accounts
• WhatsApp daily pickup alerts
• Society Green Soil Points
• Village demonstration plots</td>
<td>5. CUSTOMER SEGMENTS

• Bulk Generators (BWGs): Hotels, cloud kitchens, corporate IT canteens (Bandra/BKC)
• Gated Societies: Complexes >200 flats in Powai/Thane
• Smallholder Farmers: 14k+ growers in Palghar/Shahapur</td>
</tr>
<tr>
<td colspan="2">6. KEY RESOURCES

• 5,000 sq ft Mahape Hub
• 3-Wheeler EV Loaders + BLE Scales
• Next.js 15 PWA Platform Core
• Gemini 1.5 + ICAR Rule Engine</td>
<td colspan="2">7. CHANNELS

• Next.js 15 PWA Platform
• 3-Wheeler EV Fleet Touchpoints
• Palghar FPO Village Hubs
• Direct WhatsApp Business API</td>
</tr>
<tr>
<td colspan="2">8. COST STRUCTURE

• EV Doorstep Collection: ■2.80 / kg (power + driver)
• Bio-Processing & Inoculants: ■3.50 / kg (Bokashi + Vermi)
• Regional FPO Transport: ■1.20 / kg
• Tech & AI API Infrastructure: ■0.30 / kg
• Total Operating Cost: ■8.70 / kg produced</td>
<td colspan="3">9. REVENUE STREAMS

• B2B SWM Compliance Retainers: ■2,500–■5,000 / month / kitchen
• Agricultural Product Sales (Blended ■14.50 / kg):
- Vermicompost @ ■10/kg | Biochar @ ■24/kg
- Liquid NPK @ ■45/L | Bokashi Bran @ ■60/kg
• Voluntary Carbon Credits: ■850 / MT CO■e avoided</td>
</tr>
</table>

2. Unit Economics & Contribution Margin Analysis

Component	Cost / kg (INR)	% Realization	Operational Driver
EV Collection Logistics	■ 2.80	19.3%	Euler EV charging @ ■0.40/km, driver wage, battery.
Bio-Processing & Inoculants	■ 2.10	14.5%	Bokashi EM1 starter, red worms, pyrolysis fuel.
Facility Overhead & Labor	■ 1.40	9.7%	5,000 sq ft Mahape hub lease, power, 4 technicians.
Packaging & Bagging	■ 0.90	6.2%	50 kg woven HDPE sacks and 5 L recyclable carboys.
FPO Cluster Delivery	■ 1.20	8.3%	Bulk transport to Palghar/Shahapur village centers.
Cloud, DB & Gemini AI	■ 0.30	2.0%	Vercel serverless, Supabase PostgreSQL, Gemini.
TOTAL OPERATING COST	**■ 8.70**	**60.0%**	**All-inclusive fully loaded operating expense.**

BLENDED SELLING PRICE	**■ 14.50**	**100.0%**	**Vermi @ ■10, Biochar @ ■24, Liquid @ ■45/L.**
GROSS PROCESSING MARGIN	**■ 8.20**	**56.5%**	**Gross margin before regional farm transport.**

3. 36-Month Commercial Scaling & Breakeven Projections

Metric	Phase 1 (M 1–12)	Phase 2 (M 13–24)	Phase 3 (M 25–36)
Operational Hubs	1 Hub (Mahape Pilot)	2 Hubs (MMR/Thane)	5 Hubs (Regional Maha)
Annual Waste Diverted	900 MT / year	2,160 MT / year	5,400 MT / year
Active B2B Kitchens	60 kitchens	150 kitchens	380 kitchens
Supported Smallholders	650 farmers	1,800 farmers	5,000 farmers
Annual Gross Revenue	■ 1.02 Crores	■ 2.68 Crores	■ 6.85 Crores
Annual EBITDA	+■ 14.0 Lakhs	+■ 73.0 Lakhs	+■ 2.25 Crores
Breakeven Status	Month 14 Breakeven	Highly Profitable	High Growth Scale